MILITARY
SPECIFICATION

MIL-C-13142A(SigC)
15 March 1956
Superseding
MIL-C-13142(SigC)
3 December 1953

CRYSTAL UNIT DC-9-AJ

1. SCOPE

1.1 Scope.- This specification covers one type of quartz crystal unit designated as Crystal Unit DC-9-AJ consisting of a 1000 KC piezoelectric quartz crystal in a holder.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

NN-B-621	Boxes, Wood, Nailed and Lock-Corner.
QQ-S-571	Solder, Soft (Tin, Tin-Lead, and Lead Silver).
UU-T-111	Tape; Paper, Gummed (Sealing and Securing).
LLL-F-291	Fiberboard; Corrugated, Single Face (Flexible).
PPP-B-566	Boxes, Folding, Paperboard.

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JAN-P-106	Packaging and Packing for Overseas Shipment - Boxes; Wood, Nailed.
JAN-P-108	Packaging and Packing for Overseas Shipment Boxes, Fiberboard (V-Board and W-Board) Exterior and Interior.
MIL-P-116	Preservation, Methods of.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	Inspection Terms and Definitions.
MIL-STD-129	Marking for Shipment and Storage (Page revs. 14 and 57).

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DRAWINGS

SIGNAL CORPS

SC-B-6660 Crystal Unit DC-9-AJ

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

2.2 Other publications.- The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN TRUCKING ASSOCIATION, INCORPORATED

Motor Freight Classification Rules

(Application for copies should be addressed to the American Trucking Association, Inc., 1424 16th Street, N. W., Washington 6, D. C.)

CONSOLIDATED CLASSIFICATION COMMITTEE

Consolidated Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, Room 202, Union Station, Chicago 6, Illinois.

3. REQUIREMENTS

3.1 Preproduction samples.-

3.1.1 Quantities.- After award of contract, 30 samples or 10 percent of total number on order, whichever is less, shall be furnished for preproduction inspection and approval.

3.1.2 Fabrication and assembly.- Preproduction samples shall be fabricated by the contractor and shall meet all requirements specified for the equipment on contract. The preproduction samples shall be assembled using parts, materials, and processes that will be employed in production, and shall be fabricated and assembled in a manner similar to that to be used in production.

3.1.3 Approval and use.- Approval of preproduction samples shall not be construed as a waiver of any specified requirement. After being released to the contractor (see 3.1.4), preproduction samples to be offered as units on contract shall be refabricated by the contractor if necessary to meet specified requirements.

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3.1.4 Reference standards.- After preproduction samples have been approved and returned to the contractor's plant, they shall be kept intact in custody of the Government inspector until released by him. They shall be used as reference standards to resolve any differences of opinion regarding interpretation of requirements.

3.2 Parts, materials, and processes; general.-

3.2.1 Selection.- Parts, materials, and processes shall be such as to enable the crystal units to meet their performance and other requirements as specified herein.

3.3 Soldering.-

3.3.1 Solder.- Solder used for electrical connections shall be composition Sn60 conforming to Specification QQ-S-571.

3.3.2 Flux and cleaning agents.- No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use. Flux for soldering of electrical connections shall be rosin, rosin and alcohol, or rosin and turpentine.

3.3.3 Process.- There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting action, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the crystal unit. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.4 Cleaning.- Metal parts, after fabrication, shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect on the crystal unit. Corrosive material shall be removed completely before parts are assembled into the crystal unit.

3.5 Design and construction.- The design and construction of Crystal Unit DC-9-AJ shall be in accordance with Signal Corps Drawing SC-B-6680 and shall be such that the crystal unit will meet the performance requirements and tests of this specification.

3.5.1 Crystals.- Crystals shall be clean and free from cracks which are visible to the naked eye.

3.5.2 Crystal holder.- The crystal shall be mounted in a corrosion-proof holder which shall be evacuated or filled with a dry, inert gas, and hermetically sealed against moisture and dust.

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3.5.3 Electrodes.- Electrodes shall be made of coin silver, 18-8 stainless steel (18 percent chromium, 8 percent nickel), stainless steel type 410 or 430, monel metal, or nickel silver. The electrodes shall be free of all burrs and mechanical defects, and shall be adjusted to obtain full compliance with the electrical performance requirements of this specification and to insure long life to the crystal.

3.5.3.1 Plated crystals.- As an alternative to 3.5.3, electrodes may be replaced by a metal film on each major surface of the crystal. The plating material and process shall be in accordance with good commercial practice and the current state of the art. The material and process shall be selected and applied so as to obtain compliance with the requirements of this specification and to insure long life and stability to the crystal unit.

3.5.4 Electrical connections.- The electrical connections between the contact plates and the pins shall be made by means of soldered or welded joints, or by conductors clamped between metal surfaces. In plated type units, contact to the plating shall be made by means of contact springs or clips so designed that movement of the crystal and rubbing action between the contact and plating are reduced to a minimum.

3.5.5 Etching.- After completion of the final lapping process, quartz plates shall be finished by etching to the final frequency for unplated plates, and to the preplating frequency for metal-plated plates. The major surfaces of a plate shall not be lapped after etching, but the edges of the plate may be ground with a banded abrasive. After etching, the frequency of the plate shall have increased from the frequency measured before etching by at least the amount indicated in the following table, depending on the mesh size of the abrasive used in lapping, when the plate is tested in accordance with 4.6.11.

Mesh size	Frequency increase in cycles per second
1000	300
2000	200
3000	100
4000 or finer	40

Abrasives coarser than 1000 mesh shall not be used in the final lapping process.

3.6 Marking.- Crystal units shall be marked in accordance with Signal Corps Drawing SC-B-6680.

3.7 Performance.-

3.7.1 Frequency tolerance and temperature.- The fundamental operating frequency of Crystal Unit DC-9-AJ, shall be 1000 kc plus or minus 15 cycles at 77° ± 9° F. (25° C ± 5° C) when tested as specified in 4.6.5.2.. The fundamental frequency throughout the temperature range of -22° F to +122° F (-30° C to +50° C), shall be within plus or minus 60 cycles of its value at 77° ± 9° F. (25° ± 5° C). The crystal unit shall operate over the range of -40° F to +140° F (-40° C to +60° C).

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3.7.2 Maximum resonance resistance.- The maximum resonance resistance shall not exceed 850 ohms throughout the temperature range of -22° F to +122° F (-30° C to + 50° C).

3.8 Stability.- The stability of the crystal unit shall be such that the frequency and maximum resonance resistance shall be within the tolerance specified in 3.7.1 and 3.7.2 after each one of the tests listed in section 4 of this specification with an additional two cycles deviation from 1000 kc permitted. This additional deviation is specifically allowed to compensate for unavoidable shock, tilt, and transfer between different test oscillators.

3.8.1 Vibration, Immersion and Humidity.- The frequency of the crystal unit shall not change from the measurements prior to the tests by more than 8 cycles; the pin to pin and metal holder, DC resistance, shall not be less than that specified in 3.8.4; and the maximum resonance resistance shall not exceed 850 ohms when tested in accordance with 4.6.7, 4.6.8 or 4.6.9 and 4.6.12.

3.8.2 Tilting, Shaking and Impact.- The frequency of the crystal unit shall not change from the measurements prior to the tests by more than 5 cycles; and the maximum resonance resistance shall not exceed 850 ohms when tested in accordance with 4.6.3, 4.6.4 or 4.6.6.

3.8.3 Aging.- The frequency of the crystal unit shall not change from the measurements prior to the test by more than 10 cycles when tested in accordance with 4.6.10.

3.8.4 Insulation resistance.- The insulation resistance of the crystal unit shall not be less than 500 megohms when tested as specified in 4.6.12.

3.9 Workmanship.- The equipment shall be manufactured and assembled in a thoroughly workmanlike manner, in accordance with the applicable portions of the following paragraphs herein.

3.3.1 Solder
3.3.2 Flux and Cleaning Agent
3.3 Process for soldering
3.4 Cleaning
3.5.1 Crystals
3.5.2 Crystal holders
3.5.3 Electrodes
3.5.3.1 Plated crystals
3.5.4 Electrical connections
3.5.5 Etching
3.6 Marking

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection, general.-

4.1.1 Definitions.- Standard MIL-STD-109 applies for definitions of inspection terms used herein.

4.1.2 Classification of inspection.- Inspection shall be classified as follows:

(a) Preproduction inspection

(b) Acceptance inspection

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4.2 Preproduction inspection.- Preproduction inspection shall consist of the group A inspection and nondestructive group B and group C inspection specified in tables I, II, and III, respectively. Other nondestructive inspection on preproduction samples may be performed to determine compliance with specified requirements. Preproduction inspection will normally be performed in this order: (1) vibration, (2) bounce, and (3) immersion; other preproduction inspection may precede, follow, or be interspersed between the foregoing.

4.3 Acceptance inspection.- Acceptance inspection shall include group A, group B, and group C inspection as specified in 4.3.1 through 4.3.3.2. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. In addition, group C inspection shall normally be performed on sample units that have been subjected to and met group B inspection. However, the order may be varied when the Government considers it more practical to select separate samples for group B or group C inspection, or both.

4.3.1 Group A inspection.- Group A inspection (including sampling) shall conform to table I and Standard MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government inspector.

TABLE I - Government group A

Test	Rqt. Par.	Test Par.	AQL (per cent defective)	
			Major	Minor
Visual and mechanical inspection external	3.2 to 3.5.2 incl., 3.6 and 3.9	4.6.1	)	
Tilt	3.8.2	4.6.3	) 1.5%	6.5%
Shake	3.8.3	4.6.4	)	
Aging	3.8.3	4.6.10	1.5%	-
Temperature coefficient and activity	3.7	4.6.5	1.5%	-

4.3.2 Group B inspection.- Group B inspection (including sampling) shall conform to table II and the Appendix to Standard MIL-STD-105. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1. Disposition of rejected product (sample units and inspection lots) shall be in accordance with Standard MIL-STD-105 and 4.3.5.

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4.3.2.1 Group B sampling plans.- The group B sampling plans, as listed in table II, shall be as follows:

Group B plan	AQL	Inspection level for normal inspection	Inspection level for reduced inspection
B2	6.5%	L-7	L-5

4.3.2.2 Order of inspection within group B.- Group B inspection shall be performed in any order which is satisfactory to the Government inspector.

4.3.2.3 Procedure in case of failure.- When an inspection lot is rejected the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

TABLE II - Government Group B inspection

Test	Rqt. Par.	Test Par.	
Impact	3.8.2	4.6.6	B2
Vibration	3.8.1	4.6.7	
Immersion (when applicable)	3.8.1	4.6.8	
Humidity (when applicable)	3.8.1	4.6.9	
Visual and mechanical inspection (internal)	3.2 to 3.5.4 incl., 3.6 and 3.9	4.6.2	

4.3.3 Group C inspection.- Group C inspection shall be as listed in table III.

4.3.3.1 Sampling for inspection of equipment.- One freshly lapped plate shall be selected once each day from the production line without regard to its quality. If the sample fails, a second sample of two plates shall be selected and tested, no failing units shall be allowed in the second sample.

4.3.3.2 Noncompliance.- If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

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TABLE III. Government group C inspection

Test	Req. Par.	Test Par.
Etching	3.5.5	4.6.11

4.3.3.3 Disposition of plates.- Plates that have passed the etching test may be delivered on the contract.

4.3.4 Reinspection of conforming group B and group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

4.3.5 Disposition of nonconforming product.- When defective sample units or rejected inspection lots are resubmitted or affected portion of product is submitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The record shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.4 Test conditions.- Unless otherwise specified herein, all measurements and tests shall be made at room ambient temperature, pressure, and humidity.

4.5 Test equipment and inspection facilities.- Test equipment and inspection facilities shall be of sufficeint accuracy, quality, and quantity to permit performance of the required acceptance tests. The manufacturer shall establish adequate calibration of test equipment to the satisfaction of the Government.

4.5.1 Government reference standard test set.- The Government reference standard test shall be Crystal Impedance Meter TS-330/TSM. It shall be Government furnished for the duration of the contract, and shall be the comparison standard used for checking the calibration of the manufacturer's equivalent test sets. (see 4.5.2) The Government reference standard test set shall not be used for the performance of acceptance tests.

4.5.2 Manufacturer's equivalent test sets.- The manufacturer's equivalent test set or sets shall duplicate as nearly as possible the characteristics of the Government reference standard test set, and shall permit external connections to a crystal unit to be tested. During use, the manufacturer's sets shall be checked daily against the Government reference standard test set.

4.5.2.1 Frequency correlation.- The frequency of a crystal unit as measured in any one of the manufacturer's equivalent test sets shall not differ from the frequency of the same unit as measured in the Government reference standard test set by more than ± 0.0005 percent of the latter measurement.

4.5.2.2 Resonance resistance correlation.- The resonance resistance of a crystal unit as measured in any one of the manufacturer's equivalent test sets shall not differ from the corresponding measurement in the Government reference standard test set by more than ± 10 percent of the latter measurement.

4.6 Tests.-

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4.6.1 Visual and mechanical inspection (external).- Crystal units shall be examined externally for defects listed in table IV.

TABLE IV - Classification of Visual and Mechanical Defects (External)

Requirement	Rqt Par	Defect Classification
Materials:	3.2	
Corroded metal		Major
Design and construction:		
Outline dimensions, including pin-diameter and pin-spacing dimensions, are exceeded. (This also includes defects caused by excess solder.)	3.5, 3.5.2	Major
Other than specified material employed in holder.	3.5.2	Major
Marking:	3.6	-----
Type designation, specified frequency, mfr's code, and date of mfg. illegible.		Major
Marking not in accordance with drawing.		Major
Marking not properly positioned with respect to center, axis of crystal holder, or not properly located.		Minor
Workmanship:	3.9	-----
Missing, wrong, or defective parts.		Major
Soldering fluxes and their residues not removed after soldering.		Major
Loose solder or other foreign particles in crystal unit (check by shaking).		Major
Burrs and sharp edges not removed from metal parts.		Minor
Incorrect mounting of cover holder to base.		Minor
Loose screws, nuts, or other fastening devices.		Major
Stripped screws or screw inserts.		Major
Burred screw heads.		Minor
Gasket torn, twisted, soiled, or protruding more than 1/32 inch.		Minor
Metal soiled or dented:		Minor

4.6.2 Visual and mechanical inspection (internal).- Crystal units shall be disassembled and the interior examined for the defects listed in Table V.

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TABLE V - Classification of Visual and Mechanical Defects (Internal) 1/

Requirement	Ref. Par
Materials: Corroded parts, including suspension wires. Materials not as specified.	3.2 to 3.4 Incl
Design and construction: Design and construction not as specified.	3.5 to 3.5.5 Incl
Solder seals: All surfaces of contact between holder cover and base not joined with solder.	3.5.2
Workmanship: Unsoldered joint. Cold-solder joint (no adherence). Rosin or Flux joint. Potential short in wiring. Bonding cement loose or missing. Interior of holder cover or base contains particles of solder, metal, flux, quartz, dust, oil, wax or other foreign particles. Quartz plate oily, greasy, finger marked, or soiled by dirt or flux. Missing, wrong, or defective parts. Contact plates broken. Stripped screws. Spacers cracked or broken.	3.9

1/ All internal defects are considered major.

4.6.3 Tilt test.- After the crystal unit has been tested as specified in 4.6.5.2, 4.6.5.3 and 4.2 it shall be tilted at least 90 degrees in each of three mutually perpendicular planes and restored to normal operating position. Frequency and maximum resonance resistance readings shall be measured at 25° ± 5° C after each tilt. (See 3.8.4).

4.6.4 Shake test.- After the crystal unit has been tested as specified 4.6.5.2, 4.6.5.3 and 4.2, the crystal unit shall be removed from the oscillator, shaken violently by hand, and replaced in the test oscillator (see 3.8.2).

4.6.5 Temperature coefficient and activity.- The fundamental frequency and maximum resonance resistance shall be measured at temperature intervals not exceeding five degrees C over the temperature range of -22° F to +122° F (-30° C to +50° C). The tests shall be made in dry atmosphere to eliminate humidity effects. The rate of change of temperature shall not exceed 2° C per minute. The air in the test chamber shall be kept in high turbulence to insure uniform heating or cooling. (See 3.7.1 and 3.7.2).

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4.6.5.1 Test equipment for frequency and maximum resonance resistance.- The temperature of the crystal units under test shall be checked by means of a low thermal-mass temperature indicating device such as a thermocouple pyrometer, or resistance thermometer. The indicating device shall be placed in a dummy holder identical in type to that of the crystal units under test, and located adjacent to them. Test methods may include automatic or manual devices for checking the frequency and maximum resonance resistance of crystal units in groups at the specified intervals throughout the complete temperature range.

4.6.5.2. Frequency.- Insert crystal unit into the test set and measure the fundamental operating frequency with the crystal adjusting capacitor set at the 27 mmf point. The fundamental operating frequency shall be within the requirements of 3.7.1.

4.6.5.3 Resistance.- With the crystal unit inserted in the test set, measure the maximum resonance resistance with the drive level adjustment performed at 1.0 mc/s with 20 milliamperes of current through a 50 ohm resistor. The maximum resonance shall not exceed the requirements of 3.7.2.

4.6.6 Impact.- After the crystal unit has been tested as specified in 4.6.5.2, 4.6.5.3 and 4.2 each of the selected crystal units shall be dropped once from a height of three inches onto a hardwood surface. The crystal units may be dropped in any random position; however, this position shall not be the same for all samples (see 3.8.2).

4.6.7 Vibration.- The crystal units shall be tested as specified in 4.6.5.2, 4.6.5.3 and 4.2. With the crystal unit rigidly mounted on a horizontal platform in any position relative to the platform, simple harmonic motion of an amplitude of 0.015 inch (maximum total excursion of 0.030 inch) shall be applied to the platform. The frequency of the applied vibration shall be varied at a uniform rate between approximately 10 and 55 cycles per second; the cycle of frequency excursion shall be accomplished in not less than one minute. The vibration shall be applied continuously for two hours. (See 3.8.1) The crystal units shall then be tested for the requirements of 3.8, 3.8.1 and 3.8.4.

4.6.8 Immersion test.- After the crystal unit has been tested as specified in 4.6.5.2, 4.6.5.3 and 4.2 the crystal units shall be immersed in fresh water at room temperature. The temperature shall be gradually increased to + 60° C over a period of approximately 30 minutes and held at that point for at least 5 minutes. The temperature of the water shall then be lowered to + 10° C in approximately one hour, held at that point for at least five minutes, and then raised to room temperature in approximately one hour. The crystal units shall be removed from the water, the excess water wiped off and the units allowed to dry for two hours (max.) at room temperature. The crystal unit shall then be tested for the requirements of 3.8, 3.8.1 and 3.8.4.

4.6.9 Humidity test.- This test may, at the contractors option be substituted for the immersion test. The crystal unit shall be subjected to 90 percent relative humidity at 60° C in a chamber for a period of 24 hours. In addition to meeting previously stated requirements, there shall be no evidence of corrosion. (See 3.8.1 and 3.8.4).

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4.6.10 Aging.- The frequency of the crystal unit shall be measured as specified in 4.6.5.2, 4.6.5.3 and 4.2. The unit shall then be placed in a well ventilated oven equipped with heating and timing controls that produce the heat cycle specified in Table VI. The unit shall be subjected to three continuous full heat cycles. After this test, the crystal unit shall be tested for the requirements of 3.8 and 3.8.3.

TABLE VI - Heat cycle for aging

Step	Oven temperature: Beginning of step	Oven temperature: End of step	Maximum rate of temperature change
	°C	°C	°C/minute
1	25 +10 -5	100 ±5	2
2	Hold oven temperature constant at 100° ± 5° C for a minimum of 30 minutes		
3	100 ± 5	25 + 10 -5	2

4.6.11 Etching.- The frequency of the plate shall be measured under the conditions specified in 4.2. The plate shall then be subjected to the manufacturer's etching procedure. After the etching process, the frequency shall be measured again under the conditions specified in 4.4. (See 3.5.5).

4.6.12 Insulation resistance.- Using a 50 to 100 volt direct-current service, the resistance shall be measured between the pins of the crystal unit. For units with metal holders, measurements shall also be made between any pin and any other external metal part of the unit. The unit shall meet of requirements of 3.8.4.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 Level A - Military package.- Crystal Unit DC-9-AJ shall be preserved and packaged in accordance with the procedure specified for the designated methods as prescribed in Specification MIL-P-116 and described in the following paragraphs:

5.1.1.1 Crystal Unit DC-9-AJ.- Package each crystal unit individually in accordance with Method III as follows: Cushion each crystal unit by wrapping in flexible, single-face corrugated fiberboard, Type I, conforming to Specification LLL-F-291. Secure cushioning with Class I gummed paper tape conforming to Specification UU-T-111. Place each cushioned crystal unit within a close-fitting, folding, paperboard box, Style II, Type A, Class A conforming to Specification PPP-B-566. Secure closure with Class 2, gummed paper tape conforming to Specification UU-T-111.

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5.1.1.2 Intermediate package.- Five packaged crystal units or a multiple thereof shall be placed together within a close-fitting, regular slotted style, Grade 5 corrugated fiberboard box conforming to Specification JAN-P-108. Closure shall be in accordance with the appendix of the box specification. When specified in the bid request or contract, packages prepared as specified above, shall be required to pass the rough handling test prescribed in Specification MIL-P-116, except that they shall not be required to pass the immersion or water-spray test.

5.1.2 Level B - Limited military package.- Crystal Unit DC-9-AJ shall be packaged as specified in 5.1.1.

5.1.3 Level C - Minimum military package.- Crystal Unit DC-9-AJ shall be packaged in accordance with commercial practice and in a manner that will guarantee protection against corrosion, deterioration and physical damage during direct shipment to the first receiving activity at lowest transportation rate.

5.2 Packing.-

5.2.1 Level A - Military pack.- A quantity of Crystal Unit DC-9-AJ shall be packed in a Style 4 nailed wood box conforming to the requirements of Specification JAN-P-106. Fabricate the box to fit the contents snugly. The gross weight of any shipping container shall not exceed 150 pounds. Close the shipping container as specified in the appendix of the box specification.

5.2.1.1 Metal strapping.- Shipping containers shall be strapped in conformance with the requirements of the appendix of the container specification only for direct shipment to ports.

5.2.2 Level B - Limited military pack.- A quantity of Crystal Units DC-9-AJ shall be packed in a Style 4 nailed wood box conforming to Specification NN-B-621. Fabricate the box to fit the contents snugly. The gross weight of any shipping container shall not exceed 200 pounds. Closure shall be in accordance with the box specification. Strapping is not required.

5.2.3 Level C - Minimum military pack.- Crystal Unit DC-9-AJ shall be packed for shipment in a manner conforming to the requirements of Uniform Freight Classification for rail shipment, National Motor Freight Classification for truck shipment, Parcel Post Regulations and the regulations of other carriers as applicable to the mode of transportation employed.

5.3 Marking.- Interior packages and exterior shipping containers shall be marked in accordance with the applicable provision of Military Standard MIL-STD-129.

6. NOTES

6.1 Intended use.- This crystal unit is used as the frequency-controlling element in calibrating circuits of precision equipment.

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6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Level of packaging and level of packing required for shipment (Level A, level B or level C).

(c) Samples required (See 3.1).

(d) Marking and shipping of samples.

(e) Place of final inspection.

(f) Technical literature required.

(g) Classification of defects.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture use or sell any patented invention that may in any way be related thereto.

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